

LUNENBURG, NS, Canada

WMO: 710415

Lat: 44.360N

Lon: 64.300W

Elev: 12

StdP: 14.69

Time Zone: -4.00 (NAA)

Period: 02-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	6.7	10.8	0.4	5.6	12.2	3.5	6.6	14.5	36.5	35.0	33.0	32.3	13.7	310	0.660

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	10.5	76.3	65.9	73.6	65.0	71.2	64.4	69.3	73.7	67.7	71.1	66.3	69.2	11.9	280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
67.9	102.7	70.8	66.4	97.3	69.3	65.1	92.8	68.0	33.4	74.0	32.1	71.2	31.0	69.3	77.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
31.5	27.4	24.3	2.3	82.8	5.0	2.6	-1.4	84.7	-4.3	86.3	-7.1	87.8	-10.8	89.7		
			3.7	72.0	3.4	2.3	1.3	73.6	-0.7	74.9	-2.6	76.2	-5.0	77.8		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	46.3	27.3	27.4	32.3	41.1	49.2	57.6	64.6	65.2	60.8	52.1	42.8	34.1
	DBStd	14.80	8.85	7.89	7.02	5.21	4.84	4.88	3.72	3.58	4.97	6.14	6.79	7.85
	HDD50	3004	704	633	549	270	73	3	0	0	0	49	230	493
	HDD65	6927	1169	1053	1013	717	491	227	53	40	141	400	666	957
	CDD50	1658	0	0	1	3	48	231	451	472	325	114	13	0
	CDD65	109	0	0	0	0	1	5	39	47	16	1	0	0
	CDH74	227	0	0	0	0	6	22	96	66	36	1	0	0
	CDH80	29	0	0	0	0	0	1	17	4	7	0	0	0
Wind	WSAvg	11.2	13.0	13.0	12.9	11.7	10.4	9.8	8.3	8.3	9.6	11.6	12.3	13.2
Precipitation	PrecAvg	56.6	5.4	4.5	4.9	4.4	4.3	4.0	3.7	3.5	4.4	5.3	5.9	6.1
	PrecMax	70.3	9.6	7.5	11.0	7.0	12.6	9.8	7.4	7.9	13.2	11.3	10.2	9.8
	PrecMin	42.2	2.2	1.9	1.6	1.8	1.8	1.1	0.8	1.8	1.9	1.5	3.1	2.8
	PrecStd	6.5	1.7	1.6	2.0	1.3	2.2	1.9	1.5	1.6	2.3	2.2	2.1	1.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	48.6	47.9	56.2	59.9	72.3	77.3	82.4	78.9	78.1	70.1	59.8	54.0
		MCWB	47.2	46.5	47.4	49.3	60.4	66.1	68.5	68.1	68.6	61.3	57.6	52.9
	2%	DB	45.2	43.7	47.5	54.8	64.1	71.7	75.9	75.6	73.2	65.4	57.3	50.5
		MCWB	43.5	41.7	42.5	46.7	54.4	60.4	65.4	66.3	66.3	61.2	55.2	49.3
	5%	DB	42.3	40.5	43.9	51.5	59.5	67.8	73.0	73.1	70.1	63.2	55.1	48.3
		MCWB	40.6	38.4	40.2	46.6	52.1	59.5	64.8	65.4	64.2	60.2	53.4	47.0
	10%	DB	39.4	37.9	41.1	48.7	56.0	64.8	70.3	71.1	67.7	61.2	53.0	45.6
		MCWB	37.4	36.0	38.0	44.5	50.6	58.4	64.4	65.2	63.4	58.3	51.1	43.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	47.9	47.5	50.8	53.8	62.3	67.4	70.9	71.1	70.9	65.3	59.0	53.4
		MCDB	48.6	47.8	56.1	58.0	70.2	73.4	76.5	74.5	75.9	66.9	59.5	53.9
	2%	WB	44.1	42.8	44.5	50.3	57.5	64.2	68.8	69.4	68.1	62.8	56.0	49.7
		MCDB	45.0	43.5	46.4	52.0	61.8	69.2	73.9	73.4	71.0	64.3	56.8	50.3
	5%	WB	41.1	39.5	41.5	47.9	54.2	61.9	67.1	67.9	66.3	61.2	53.7	47.5
		MCDB	42.3	40.4	43.8	50.2	57.3	65.0	70.8	70.9	68.4	62.9	54.6	48.3
	10%	WB	38.0	36.7	38.8	45.5	51.8	60.0	65.6	66.4	64.7	59.0	51.4	44.4
		MCDB	39.6	38.1	41.2	48.0	55.0	62.9	68.7	69.2	66.7	60.7	52.8	45.6
Mean Daily Temperature Range	5% DB	MDBR	11.6	11.3	10.8	10.2	10.5	10.5	9.9	10.5	10.9	10.5	10.1	10.7
		MCDBR	14.0	13.2	13.8	15.4	17.1	16.8	14.3	13.5	13.1	12.2	11.9	13.3
		MCWBR	14.7	13.1	10.5	10.1	9.9	8.9	7.2	6.9	8.4	9.4	11.8	14.6
	5% WB	MCDBR	13.7	13.5	13.3	12.6	14.3	13.9	11.3	11.1	10.0	10.6	11.6	13.1
		MCWBR	14.6	14.2	12.0	10.6	10.2	9.0	7.5	6.9	7.8	9.7	12.4	14.4
Clear-Sky Solar Irradiance	taub	0.303	0.300	0.331	0.357	0.376	0.404	0.422	0.399	0.361	0.337	0.319	0.303	
	taud	2.482	2.519	2.455	2.427	2.418	2.369	2.333	2.397	2.472	2.538	2.541	2.518	
	Ebn at Noon	261	285	288	288	284	275	268	271	274	267	254	250	
	Edh at Noon	22	25	31	35	36	38	39	35	30	25	21	20	
All-Sky Solar Radiation	RadAvg	456	732	1114	1374	1581	1734	1729	1601	1302	842	482	360	
	RadStd	34	56	85	124	200	165	123	108	82	92	52	29	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (31 neighbors)	N/A	N/A	N/A	N/A	N/A	+0.77	+1.03	N/A	N/A	+104

Nomenclature: See separate page